

SPOKANE FELTS FIELD, WA, USA

WMO: 727856

Lat: 47.683N

Lon: 117.321W

Elev: 1953

StdP: 13.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94176

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.4	15.0	-0.1	5.9	13.4	5.8	8.0	17.8	24.2	41.7	20.3	40.1	2.8	60	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.9	94.7	64.0	91.3	63.1	88.2	62.2	66.7	88.9	65.0	86.5	63.3	83.9	6.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.1	80.4	71.5	56.9	74.1	70.9	54.9	69.1	69.4	32.2	88.6	30.8	86.4	29.6	83.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	17.2	14.7	3.4	100.3	6.7	3.0	-1.5	102.5	-5.4	104.2	-9.2	105.8	-14.1	108.0
			2.7	70.8	6.6	4.3	-2.1	73.8	-5.9	76.3	-9.6	78.7	-14.4	81.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	31.8	34.3	41.3	47.8	56.7	63.0	71.6	70.4	61.4	48.8	38.6	31.2
	DBStd	15.62	8.04	7.14	6.16	6.01	6.93	6.45	5.75	5.58	6.59	6.66	7.54	7.69
	HDD50	2445	565	441	273	112	18	1	0	0	4	99	348	584
	HDD65	6042	1030	860	734	516	271	112	14	17	145	502	793	1048
	CDD50	2384	0	0	4	47	225	391	671	632	346	63	5	0
	CDD65	508	0	0	0	0	13	52	220	185	37	1	0	0
	CDH74	7504	0	0	0	23	296	831	3061	2587	685	21	0	0
	CDH80	3372	0	0	0	2	76	295	1522	1226	248	3	0	0
(14) Wind	WSAvg	4.9	3.9	4.3	5.8	6.1	5.8	5.9	5.5	5.0	4.5	4.4	4.3	3.9
(15) Precipitation	PrecAvg	23.9	2.8	1.8	2.8	1.8	1.9	1.9	0.6	0.7	1.1	2.1	3.0	3.3
	PrecMax	28.3	6.5	2.4	4.7	3.0	3.0	4.2	1.2	2.1	2.5	6.9	5.8	5.1
	PrecMin	20.1	1.4	0.6	1.5	1.0	1.1	0.6	0.1	0.1	0.4	0.6	1.8	1.5
	PrecStd	2.5	1.3	0.4	0.9	0.5	0.7	0.9	0.4	0.6	0.5	1.4	0.9	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.4	54.3	65.4	77.3	86.9	94.1	99.4	98.7	92.6	76.6	60.7	53.9
		MCWB	45.0	45.7	50.1	56.9	60.9	65.3	67.0	66.0	62.2	57.5	54.1	47.5
	2%	DB	47.1	50.1	60.3	70.9	81.6	87.6	95.4	94.3	87.0	69.7	55.6	47.6
		MCWB	42.3	43.5	48.1	53.7	59.6	62.6	64.3	63.7	61.0	54.4	49.0	43.4
	5%	DB	44.5	46.9	56.1	65.1	76.8	82.4	92.2	90.9	81.7	64.3	52.4	44.4
		MCWB	40.4	41.1	45.8	50.5	56.7	60.7	63.5	62.8	59.7	52.0	46.7	40.6
	10%	DB	41.5	44.4	52.4	60.8	71.9	78.7	89.1	87.6	77.0	60.6	49.3	41.0
		MCWB	38.6	39.2	44.1	48.0	54.8	58.6	62.7	62.0	57.4	50.0	44.6	38.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	47.3	52.2	58.5	63.7	67.1	69.9	69.0	65.4	59.3	54.6	47.9
		MCDB	49.2	52.4	61.2	74.1	81.1	89.1	94.6	92.7	84.6	73.1	59.1	53.1
	2%	WB	43.0	44.4	49.4	54.5	60.3	64.2	67.2	66.2	62.6	55.9	50.3	44.0
		MCDB	46.3	49.0	58.0	68.5	77.3	83.7	89.2	89.2	82.4	66.1	54.8	47.6
	5%	WB	40.8	42.0	47.1	51.7	58.4	62.0	65.5	64.2	60.4	53.4	47.7	41.0
		MCDB	43.8	46.1	54.5	63.5	73.5	79.9	87.5	86.1	78.5	62.3	51.3	43.9
	10%	WB	38.8	39.8	44.9	49.4	56.2	59.9	63.7	62.4	58.4	51.1	45.1	38.4
		MCDB	41.5	43.7	51.7	59.0	69.3	75.6	84.8	83.5	74.4	58.5	48.3	40.8
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	13.7	17.9	20.7	23.2	23.4	28.9	29.2	26.4	20.1	12.9	9.7
		MCDBR	12.8	17.5	24.6	29.6	31.7	31.8	35.1	36.1	35.1	27.1	16.1	12.5
		MCWBR	9.1	11.4	14.2	14.8	13.0	11.6	11.3	12.6	13.6	13.7	10.6	9.4
	5% WB	MCDBR	12.1	15.2	21.7	27.4	28.3	29.6	32.6	32.9	31.8	23.8	13.6	11.9
		MCWBR	9.0	10.4	13.5	14.4	12.2	11.3	11.4	12.4	13.0	12.6	9.6	9.5
(40) Clear-Sky Solar Irradiance	taub		0.277	0.287	0.307	0.316	0.330	0.333	0.333	0.346	0.321	0.307	0.287	0.275
	taud		2.483	2.512	2.518	2.489	2.473	2.497	2.495	2.446	2.506	2.540	2.546	2.483
	Ebn at Noon		260	283	292	298	296	295	293	284	283	271	255	246
	Edh at Noon		20	24	28	32	33	33	32	33	28	23	19	18
(44) All-Sky Solar Radiation	RadAvg		328	621	952	1428	1839	1976	2333	1937	1378	815	386	258
	RadStd		42	85	114	138	148	175	131	129	132	93	43	34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.24	N/A	N/A	N/A	+203	+79
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	N/A

Nomenclature: See separate page